
Lecture Notes

on Partition Theorem
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1 Partition Theorem

1.1 Definition:

Let S be a sample space and $B_1, B_2, \dots, B_n$ be a sequence of **mutually exclusive** events such that

$$B_1 \cup B_2 \cup \dots \cup B_n = S, \quad \text{and} \quad B_i \cap B_j = \emptyset \text{ for } i \neq j.$$

Then the collection $\{B_1, B_2, \dots, B_n\}$ is called a **partition of the sample space** S .

1.2 Partition Property: (Total probability).

Let $\{B_1, B_2, \dots, B_n\}$ be a partition of a sample space S . Then, for any event A in S , the probability of A is given by:

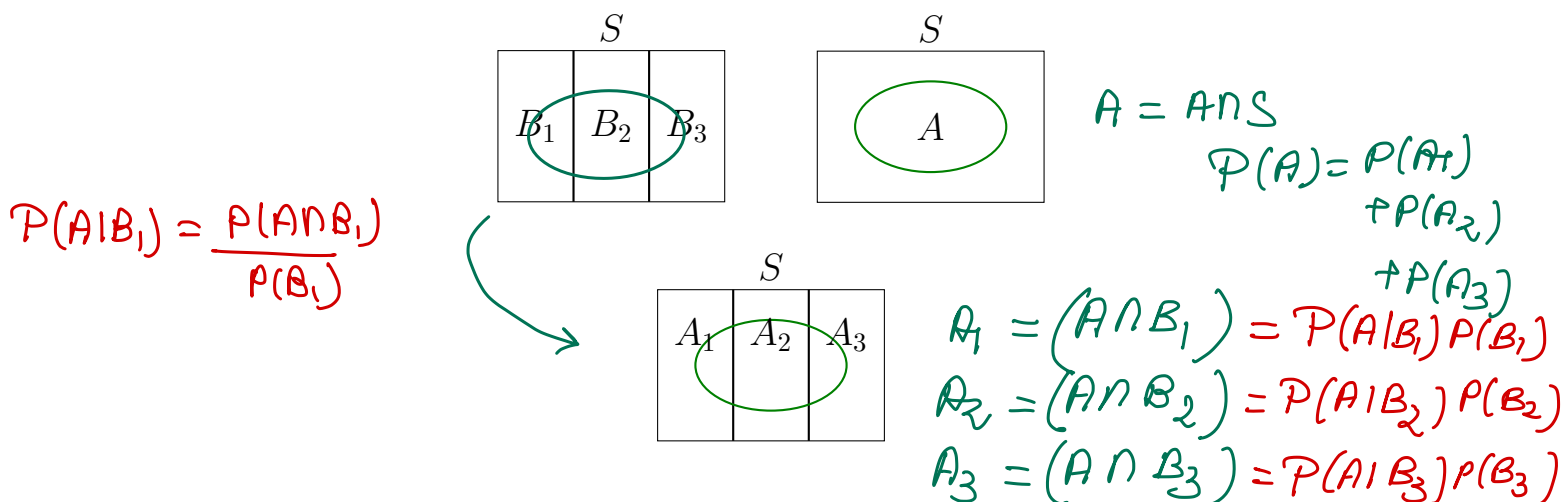
$$P(A) = \sum_{i=1}^n P(A | B_i) P(B_i)$$

Proof:

$$\begin{aligned} P(A) &= P(A \cap S) \\ &= P(A \cap (B_1 \cup B_2 \cup \dots \cup B_n)) = P((A \cap B_1) \cup (A \cap B_2) \cup \dots \cup (A \cap B_n)) \\ &= P(A \cap B_1) + P(A \cap B_2) + \dots + P(A \cap B_n) \\ &= P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_n)P(B_n) \\ &= \sum_{i=1}^n P(A|B_i) P(B_i). \end{aligned}$$

$\therefore P(A \cup B) = P(A) + P(B)$

This result is known as the **Partition Theorem**, also called the **Total Probability Theorem**.



$$P(A) = ?$$

$$P(A) = P(A|H)P(H) + P(A|T)P(T)$$

☑ **Example** A fair coin is flipped. If a head turns up, a fair die is tossed; if a tail turns up, two fair dice are tossed. What is the probability of the event A that the sum of the appearing number(s) is equal to 6?

Here $S = \{H, T\}$. Let H denotes the event of having head and T denotes the event of having tail.

$$P(H) = \frac{1}{2} = P(T).$$

$$P(A|H) = \frac{1}{6}.$$

$$P(A|T) = \left(\frac{5}{36}\right).$$

$$A_1 = \{6\}$$

$$S_1 = \{1, 2, 3, 4, 5, 6\}$$

$$A_2 = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$$

$$S_2 = \{(1, 1), (1, 2), \dots, (6, 6)\}$$

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$$\therefore P(A) = P(A|H)P(H) + P(A|T)P(T)$$

$$= \left(\frac{1}{6} \times \frac{1}{2}\right) + \left(\frac{5}{36} \times \frac{1}{2}\right)$$

$$= \frac{1}{12} + \frac{5}{72} = \frac{6+5}{72} = \frac{11}{72}.$$

✎ **Example** If men constitute 47% of the population and tell the truth 78% of the time, while women tell the truth 63% of the time, what is the probability that a person selected at random will answer a question truthfully?

Let A denote the event of the person answer truthfully.
Let M " " " " " " " " " " to be men.
" W " " " " " " " " " " women.

$$P(A) = ?$$

$$P(A) = P(A|M)P(M) + P(A|W)P(W).$$

$$P(M) = 0.47$$

$$P(W) = 1 - 0.47 = 0.53.$$

$$P(A|M) = 0.78$$

$$P(A|W) = 0.63.$$

$$\begin{aligned}\therefore P(A) &= P(A|M)P(M) + P(A|W)P(W) \\ &= (0.78 \times 0.47) + (0.53 \times 0.63). \\ &\approx 0.70.\end{aligned}$$

✎ **Example** In a certain city, 40% of the population takes public transportation to work, 35% drives their own car, and 25% works from home. The probability of being late to work is 0.10 for those who take public transportation, 0.05 for those who drive, and 0 for those who work from home. What is the probability that a randomly selected person from this city is late to work?

Let A denote the event of a randomly selected person is late to work.

B_1 = The event of using the public transportation to work.

B_2 = " " " " own car

B_3 = " " of working from home.

$P(A) = ?$

Given, $P(B_1) = 0.4$, $P(B_2) = 0.35$ and $P(B_3) = 0.25$.
 $P(A|B_1) = 0.10$, $P(A|B_2) = 0.05$ and $P(A|B_3) = 0$.

∴ By using partition theorem,

$$\begin{aligned} P(A) &= P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + P(A|B_3)P(B_3) \\ &= (0.10 \times 0.4) + (0.05 \times 0.35) + (0 \times 0.25) \\ &= 0.04 + 0.0175 + 0 \\ &= 0.0575 \end{aligned}$$